Heritability and social interaction in labor market outcomes: Evidence from the population of twins in the United States

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Abstract

We use data on 44,000 twin pairs observed as adolescents in the 2000 Census, linked to their earnings 20 years later, to study the heritability of labor market outcomes. We extend the Classical Twins Design (CTD), which identifies heritability from contrasts between monozygotic and dizygotic twins, to settings where an analyst does not observe zygosity but can measure outcomes for same-sex and opposite-sex twins. We also allow for the presence of a family- and sex-specific component affecting same-sex siblings but only partly shared by opposite-sex siblings. Our extended model identifies heritability from the difference-in-differences of same-sex vs. opposite-sex twins vs. siblings. The estimates indicate strong heritability (h2 » 0.36) of log earnings. Using an AKM (Abowd et al., 1999) decomposition that separates the person-specific component of earnings from employer-specific pay premiums, we find that both components are heritable. The heritability of employer pay premiums, however, is mainly driven by same-sex twins who work at the same firm, suggesting that social interactions may lead to an over-estimate of the genetic component in earnings. Excluding siblings who work together leads to estimates of the heritability of log earnings that are 15% lower. Similar biases may be present in CTD-based estimates of heritability for other social outcomes like education.

JEL Classification: C31, D31, J24, J31, J62

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Introduction

Social scientists have long been interested in the intergenerational transmission of economic status, as measured by occupation (Blau and Duncan, 1967; Jencks, et al., 1979; Goldthorpe, et al., 1980; Hout, 1988), earnings (Behrman and Taubman, 1985; Becker and Tomes, 1986; Zimmerman, 1992), or income (Treiman and Hauser, 1977; Solon, 1992). Part of the correlation in economic status between parents and children is due to the effect of parents' resources and social status. Part is also due to genetic inheritance. Typical data sources do not allow a clear distinction between these channels.

Twins studies offer a solution to this problem, making it possible to measure the heritability of physical traits like height (Silventoinen et al. 2003) as well as of social outcomes like IQ (Plomin, Owen, and McGuffin, 1994), education (Bingley, Cappellari, and Tatsiramos, 2024), and income (Taubman, 1976; Björklund, Jäntti, and Solon, 2005; Benjamin et al., 2012; Sacerdote 2011).[1] Monozygotic (identical; MZ) twins share all of their genes, while dizygotic (fraternal; DZ) twins share the same fraction as other siblings. Both types of twin pairs share a family environment. In the standard ACE model (also known as the "classical twin design" or "CTD" – see, e.g., Visscher 2004), the heritability of a trait, h^2, defined as the share of the variance of the trait attributable to genes, is measured as twice the difference between the MZ and DZ correlations of that trait. Silventoinen et al. (2003) conduct twins studies of height using data from seven countries, finding values of h^2 from 0.68 to 0.85. A more recent meta-analysis of twins studies of height at age 17 (Dewau et al., 2025) finds mean h^2 of 0.77 for boys and 0.62 for girls. Polderman et al. (2015) conduct a meta-analysis of 2,748 twins studies, classifying them by the outcome domain. Most studies relate to physical and mental health domains, but Polderman et al. (2015) identify 292 studies of cognitive traits (average h^2=0.468; SE=0.010), 56

[1] A separate social sciences literature uses differences between siblings or twins to identify the causal return to education, on the grounds that twins have identical ability and thus differences between them removes omitted variables bias (Griliches, 1979; Ashenfelter and Krueger, 1994; Card, 1999). Other literatures use adoptees (e.g., Sacerdote 2011; Björkland, Lindahl, and Plug 2006; Black et al. 2020) or direct genetic measures (see the review by Benjamin et al., 2024), rather than twins, to identify heritability.

studies of interpersonal interactions and relationships (average h^2=0.319; SE=0.038), and 35 of social values (average h^2=0.313; SE=0.019).

Unlike genomic studies that use direct measures of genotypes, and thus must rely on the selective samples and limited outcomes available in biobanks (e.g., Schoeler et al., 2023), twins studies do not use genetic information. Standard implementations do, however, rely on the distinction between MZ and DZ twin pairs -- information that is usually unavailable in population data. As a result, most twin studies use twins who happen to be included in specialized surveys (e.g., the Add Health data). Samples are often very small.

We report twins correlations for log earnings based on the full population of twins from eight birth cohorts (born 1983 to 1990) in the United States (N=44,500 twin pairs[2]). We use administrative earnings data that lack information on zygosity. We can, however, distinguish same-sex (SS) from opposite-sex (OS) twins. All OS twins are DZ, while SS twins are a mix of MZ and DZ, with estimable proportions. These data are sufficient to identify h^2 under standard CTD assumptions.

If boys and girls are treated differently in some families, however, SS twin correlations will be higher than OS correlations, biasing estimates of h^2 based on the standard ACE assumptions. We extend the model to allow for a family-by-sex component, which we identify by comparing SS to OS non-twin siblings. The SS-OS gap in sibling correlations is much larger for twins than for non-twin siblings – even those very close in age. In our extended ACE model, we identify h^2 from the difference-in-differences of these correlations -- SS minus OS twins vs. SS minus OS siblings. Using this approach, we estimate substantial heritability in the probability of employment (h^2=0.23-0.32; SE=0.05) and in log earnings conditional on working (h^2=0.36; SE=0.05).

There have been a few twins-based estimates of the heritability of labor market outcomes. Our samples are much larger and more representative than others in the existing literature (e.g., Bingley, et al., 2024). Nevertheless, our specific h^2 estimates have to be interpreted carefully, as they rely on the assumptions of our extended ACE (E-ACE) model.

[2] Our analyses rely on confidential data maintained by the U.S. Census Bureau. Sample sizes are rounded to the nearest 100.

Specifically, the model assumes that the family environment is equally shared by MZ and DZ twins, and by SS and OS twins, net of a gender difference that is *the same* for SS twins as for non-twin SS siblings but potentially different for girls and boys in the same family (Gusev, 2024b, 2025; Bingley, et al., 2024; Burt and Simons, 2014). We present evidence in support of this assumption. Insofar as some of the shared component of earnings for identical twins is attributable to direct interactions between the twins, however, (e.g., Kaprio, Koskenvuo, and Rose, 1990), our E-ACE model will overstate h^2.

A burgeoning literature in labor economics decomposes log earnings into "person" and "firm" components, using what is known as the AKM model (Abowd, Kramarz, and Margolis, 1999; see also Kline, 2024). The "person effect" in an AKM model is the component of earnings that remains constant as a person switches between jobs. It is often interpreted as reflecting a person's human capital -- broadly defined to include personality traits, ambition, etc. The "firm effect" is the earnings premium or discount that is shared by all workers at a given firm, and is often interpreted as reflecting the firms' pay policies. Neoclassical economic models of the labor market assume there are no firm-specific pay premiums. A growing body of studies, however, find evidence of such premiums, accounting for 10-20% of the variation in wages (Kline, 2024).

Of great interest is understanding which workers obtain jobs at high-paying firms, and why (e.g., Card, Rothstein, and Yi 2024b, 2026; Gerard et al., 2021). Does access to high-paying jobs represent pure luck (Becker and Tomes, 1979), or is it related to worker skills or attributes? We use the AKM model to decompose earnings into individual and firm effects, and estimate the heritability of each component. We expect the heritability of earnings to be largely driven by heritability of the person component. Firm premiums could also be partly heritable, to the extent that firms that offer higher premiums tend to hire workers with high person effects (so-called "assortative matching"). Existing evidence on the correlation between person effects and firm premiums, however, suggests that this channel would lead to limited heritability in firm premiums—25% or less of the heritability of person effects.[3]

[3] Existing studies find that the correlation of worker and firm effects in earnings in the U.S. labor market is 0.50 or less (see e.g., Lachowska et al., 2026). This implies that the part of the firm effect that is

Estimates from our E-ACE model imply strong heritability of person effects (h^2=0.40; SE=0.05) and moderate heritability of firm effects (h^2=0.21; SE=0.08). The latter is about 2× larger than could be explained by assortative matching alone. We find heritability of firm effects even within regions and industries, suggesting that it is not solely due to inherited preferences over geography or type of work. On its face, this seems to suggest that genetic factors affect not only workers' skills but also their access to higher-paying firms.

Economic outcomes like earnings are different from physical outcomes like height or IQ, in that they depend on social interactions. For example, whether an individual takes a particular job depends on how much they value specific features of the job – such as who else works there. Social interactions could inflate the correlation between MZ twins relative to the correlation of siblings (Horwitz et al. 2003, Stenberg 2011). Suppose, for example, that identical twins have a stronger preference to work together than do same-sex siblings. Because people working at the same firm necessarily have the same firm effects, this will lead to an upward bias in the estimated heritability of firm-specific pay premiums from the E-ACE model. In our data, fully 11.4% of SS twin pairs are observed working for the same firm, nearly twice the rate for non-twin SS siblings (5.7%) and 3× or more the rate for OS twins (3.8%) or OS siblings (2.2%).

In the language of Manski (1993), the earnings components of the ACE model are "correlated effects" – determinants of the outcomes of siblings that reflect the influence of unobserved but common genes or a common family environment. But there may also be what Manski identified as "endogenous effects," where one child's outcome directly influences that of his or her sibling. We suspect that the decision to work at the same firm as one's twin could be driven in part by such endogenous social effects.

To address the potential biases in the components of our E-ACE model induced by endogenous interactions between twins, we re-estimate the model using a sample that excludes sibling pairs who work at the same firm. The estimated heritability of firm effects falls substantially and is no longer statistically distinguishable from zero (h^2=0.07; SE=0.08) or from a value that is one-quarter of the heritability of person effects (i.e., 0.35÷4=0.09). The estimated

attributable to the worker abilities and traits embedded in the person effect explain 25% or less of the variance in firm premiums.

heritability of earnings as a whole also falls by about 15% (h^2=0.31; SE=0.08). This points to the potential importance of endogenous interactions among twins (e.g., choosing to work at the same job, or to attend the same college) that can lead to inflated estimates of the heritability of social outcomes from a CTD approach. Similar biases could arise in CTD studies of heritability of outcomes like education, which could also be influenced by interactions among twins. As our "different firms" strategy eliminates only one form of endogenous interaction, we view our heritability estimates for overall earnings and for AKM person effects as upper bounds, potentially biased by uncontrolled interactions among SS twins. Such biases could be avoided in designs that use individual data on genotypes, such as polygenetic index designs (Benjamin et al., 2024). Thus, our analysis points to a potential source of the "missing heritability" problem (Maher 2008), whereby polygenic studies estimate lower heritability than do twin studies.

Results

Our linked data set, described in the methods section, includes 44,500 twin pairs with sufficient labor market attachment to measure wages. Of these, 31,500 are SS and 13,000 are OS. The data also include 507,000 SS sibling pairs and 500,000 OS sibling pairs. Among children (including non-twins) born in the same cohorts and observed at the same dates, 50.8% are male. We take that as an estimate of the fraction male among all births meeting the conditions for inclusion in our sample (including survival to adulthood and having sufficient labor market attachment). The implied share of SS twins who are MZ is s=58.7%.

Figure 1 presents the correlations of various outcomes between SS twins (ρ_{SS}), OS twins (ρ_{OS}), SS siblings (ρ_{SSsib}), and OS siblings (ρ_{OSsib}). The first panel shows the correlations of log average quarterly earnings for the 4 groups. This is 0.543 (SE=0.006) for SS twins, but much lower for the other comparisons. We note that the estimated correlation in earnings for same sex siblings – 0.386 (SE 0.005) – is very close to the average of the correlations for non-twin brothers (0.430) and non-twin sisters (0.323) reported by Mazumder (2008). SS siblings have a stronger correlation of earnings than OS siblings, consistent with the presence of a gender-specific family factor in our E-ACE model, but the SS-OS difference is much larger for twins. The difference-in-differences is $\left((\rho_{SS} - \rho_{OS}) - \left(\rho_{SS}^{sib} - \rho_{OS}^{sib}\right)\right)$=0.105 (SE=0.014). Under the

baseline assumption of our E-ACE model developed in the methods section – that siblings share one-half of their genes -- this implies h^2=0.36 (SE=0.05).

The next two panels separate mean log earnings into person effect and firm effect components. In each case, ρ_{SS} is much larger than the other three correlations, and the difference-in-differences is positive – more so for the person effect (h^2=0.40, SE=0.04) than for the firm effect (h^2=0.21, SE=0.08). The remainder of the figure decomposes the firm effects into three components following Card, Rothstein, and Yi (2024a): A location premium, defined as the average firm effect among all firms in the commuting zone; an industry premium, defined as the average firm effect among all firms in the industry; and a remainder. All three are correlated among twins and siblings, in each case more so for SS twins than for other pairs.

These estimates all condition on significant labor force attachment – pairs in which either sibling does not work (or only works infrequently) are excluded. In this regard the similarity of our estimate of the correlation in earnings between same-sex non-twin siblings with the estimates presented by Mazumder (2008), which are based on a broader sample, is reassuring. The remaining panels in Figure 1 present sibling and twin correlations of three measures of employment status, estimated over the full population of sibling pairs. All show modest positive correlations among siblings, stronger for SS twins than for any of the other groups. We also investigated the stability of the estimated correlations in Figure 1 across different ages (Bingley and Cappellari, 2019) and found very similar estimates for siblings in the 8-year age range in our data.

Table 1 presents the correlations plotted in Figure 1, along with differences between SS and OS pairs, the difference-in-differences, and the implied heritability. All outcomes show positive heritability, with larger estimates for log earnings and the person effects than for the other components. All except the location premium, which is imprecisely estimated due to the large share of children who remain in the CZ where they grew up, are significantly different from zero. Heritability estimates are plotted in **Figure 2**.

Figure 3 presents correlations computed over subsamples of siblings separated by different gaps in age, ranging from one year to three years. Among non-twin siblings the correlations decay only slightly as the age gap grows. This suggests that common family

environmental influences on earnings (the C component of the ACE model) do not fade dramatically as the age gap between siblings rises. Moreover, the SS and OS correlations decay at similar rates, supporting our modeling assumption that there is a stable family-by-gender component in earnings. We do not reject the hypothesis that the SS-OS difference in log earnings correlations is equal across all age gaps from 4 to 12 quarters (χ^2=4.21, d.f.=8, p=0.84).

When we consider twins (shown in Figure 3 with an age gap of 0), we see higher SS correlations for nearly all outcomes than for SS siblings who are even one year apart. In contrast, the difference in correlations between OS twins and OS siblings is very small, as predicted by our E-ACE model. This buttresses our interpretation that the difference-in-differences filters out both the overall family component in earnings *and* any family-by-gender component in the $(\rho_{SS} - \rho_{OS})$ gap, isolating the component due to differences in the genetic similarity of SS versus OS twins. We note, however, that if the family-by-gender component affecting same sex twins is larger than that affecting same sex siblings even 1 year apart, our approach will over-state heritability.

Table 2 presents estimates of h^2 for two samples. The first column repeats results for our main sample from Table 1. Column 2 excludes twin and sibling pairs who work at the same firm. The heritability estimate for the AKM person effect falls by about 15% while the estimate for the firm effect falls by two-thirds, and is no longer statistically significant (h^2=0.07, SE=0.08). The relative size of the heritability shares of the two earnings components in column 2 is in the range that would be predicted if *all* the genetic influence on the firm component was driven by the tendency of higher-paying firms to hire workers with higher person effects.

Discussion

Interpretation of CTD estimates is complex, especially for social outcomes. The estimated heritability of outcomes may be biased in the presence of assortative mating, genetic dominance, population stratification, or violations of the assumption that the home environment is shared to the same degree for MZ as for DZ twins. Moreover, even when estimated without bias, heritability need not indicate direct genetic effects – it also may reflect indirect or cultural effects (e.g., effects of the parents' genes on a child's outcomes, not operating through the child's own genes; see, e.g., Gusev (2024a; 2025), Howe et al. (2022)). In

the case of outcomes like earnings, another channel arises that may not be relevant for a purely physical outcome like height: Twins may influence each other directly – for example in their choice of jobs -- inflating the correlation in their outcomes.

Using our overall data set of twins and siblings we find that labor market outcomes are heritable, with estimated heritability of 0.40 for the person effect component of earnings and 0.36 for earnings as a whole. Heritability is directly related to intergenerational transmission: If a fraction h^2 of the variation in an outcome is genetic, one would expect an intergenerational correlation between a child and one of his or her parents of $0.5h^2$ due to direct genetic influences alone.[4] Thus, our estimates imply an intergenerational elasticity of earnings of 0.15-0.18 between fathers and sons, for example, absent any environmental components of transmission. This is less than one-half of the typical estimates of the IGE (e.g., Card, Rothstein, and Yi 2026), suggesting that the intergenerational transmission of earnings reflects substantial contributions from both genetic and non-genetic mechanisms.

Heritability of the firm effect component of earnings is somewhat smaller (0.21) but still surprisingly high. When we exclude sibling pairs working at the same firm to limit the effect of social interactions between twins, however, the estimated heritability of the firm component falls substantially and is no longer statistically significant.

There are non-genetic reasons for twins to tend to work at the same firm. Thus, we interpret the estimates excluding same-firm pairs as the best estimates of heritability. These indicate that log wages and the person effect component are strongly heritable (h^2=0.31 and 0.35 respectively) but that the firm effect (h^2=0.07) is not. The fall in the heritability of firm effects when twins who work for the same firm are excluded also raises a caution about potential confounding of the CTD when applied to other outcomes where individual choice plays a role. One can easily imagine similar mechanisms affecting the education, industry, or location choices of SS twins, arising from a desire to stay together rather than from any shared genetic factors. Such interactions could inflate the correlations in the person effects of same

[4] In the presence of assortative mating, the genetic contribution to the intergenerational correlation would be larger. But plausible estimates of the magnitude of assortative mating (e.g., Hugh-Jones et al., 2016; see further discussion in the methods section below) are not large enough to change this much.

sex twins, implying that even our estimates derived from the sample excluding siblings who work together may be inflated by unobserved social interaction effects.

MATERIALS AND METHODS

Data

Our analysis is based on children identified in the U.S. 2000 population census who were the biological child of the household head. We identify twins as children in the same household with the same birthdate (±2 days), and siblings as children with different birthdates (nine months or more apart). The U.S. Census Bureau assigns Personal Identification Keys (PIKs) to all respondents using name and birthdate information. We limit attention to twin or sibling pairs in which both children were assigned PIKs and were between 9 and 17 years old at the time of the census. Families with 3 or more children can contribute multiple pairs to the sample.

These individuals are linked to the Longitudinal Employer Household Dynamics (LEHD) data, which reports the quarterly earnings received by employees from each employer. LEHD data are derived from unemployment insurance tax payments, and cover all formal sector wage and salary jobs (i.e., excluding self-employment) apart from the federal government. We use LEHD data from 2018 to the first quarter of 2023, excluding the second, third, and fourth quarters of 2020 when the labor market was substantially disrupted by the COVID pandemic. We focus on individuals who are observed in the LEHD in this period with at least 8 quarters of observed earnings above $3,800 (approximately the minimum wage for full-time work), excluding the first and last quarter of each employment spell and any quarter with multiple employers. We interpret observed quarterly earnings for this sample as reflecting full-quarter, full-time earnings. In our analysis of earnings outcomes, we retain only sibling pairs in which both siblings meet the restrictions. Our primary outcome is average log earnings for the individual across all quarters meeting the above restrictions.

We also use 3 measures of employment: an indicator for having any employment during the sample period; the number of quarters with positive employment, and an indicator for having sufficient employment to enter into our sample. These are measured over the full population of sibling pairs, without conditioning on earnings.

We classify two siblings as working at the same firm if they had at least one employer in common in our 2018-2022 data window.[5] When we examine correlations for siblings observed with different age gaps, we exclude non-twin sibling pairs with gaps smaller than one year, as these likely reflect adoptions or measurement error rather than actual sibling pairs.

Extended ACE Model

Let cov_g represent the covariance of long-term outcome Y_i between siblings of type g, where the types to be considered are MZ twins ($g = MZ$); same-sex DZ twins ($g = DZSS$); all same-sex twins, combining MZ and DZSS ($g = SS$); opposite-sex twins, who are all DZ ($g = OS$); same-sex, non-twin siblings ($g = SSsib$); and opposite-sex, non-twin siblings ($g = OSsib$). We can measure cov_{SS}, cov_{OS}, cov_{SSsib}, and cov_{OSsib} directly.

In the basic ACE model (e.g., Visscher 2004), Y_i is the sum of 3 factors: genetic endowment A_i, shared family environment $C_{f(i)}$, where $f(i)$ is an index function pointing to the family of person i, and E_i, represents non-shared environment: $Y_i = A_i + C_{f(i)} + E_i$. The heritability of the outcome is defined as $h^2 = V(A)/V(Y)$, the share of the variation that is explained by the genetic endowment.

MZ twins *i* and *j* are assumed to have identical genetic endowment $A_i = A_j$. DZ twins are assumed to have correlated genetic endowment $R_A = corr(A_i, A_j)$, with $R_A < 1$. As in other CTD studies, we assume $R_A = 0.5$, as would arise under random mating, while acknowledging that with assortative mating R_A might be larger.

Both MZ and DZ twins have identical shared environments, $C_{f(i)} = C_{f(j)}$, but non-shared environments that are assumed to be independently distributed across the relevant population. Assuming variances and the A-C covariance are the same for MZ and DZ twins, $cov_{MZ} = V(A) + V(C) + 2cov(A, C)$ while $cov_{DZ} = R_A V(A) + V(C) + 2cov(A, C)$. Thus, heritability can be estimated as $h^2 = \frac{cov_{MZ} - cov_{DZ}}{(1-R_A)V(Y)} = 2(\rho_{MZ} - \rho_{DZ})$ when $R_A = 0.5$.

We cannot distinguish MZ from DZSS twins, only SS from OS. The SS twin covariance is

[5] The LEHD assigns workers to specific establishments rather than to firms (e.g., McDonalds). All references to "firms" in fact refer to specific establishments. Note that with our definition siblings may be counted as working at the same firm even if their employment spells did not overlap.

$$cov_{SS} = (s + (1-s)R_A)V(A) + V(C) + 2cov(A,C)$$

where $s \equiv p(MZ \mid SS)$ is the share of SS twins who are MZ. The OS twin covariance is the same as the DZ covariance,

$$cov_{OS} = R_A V(A) + V(C) + 2cov(A,C).$$

This implies $h^2 = (cov_{SS} - cov_{OS})/s(1-R_A)V(Y)$. Thus, the CTD can be extended directly to data where zygosity is not identified, so long as an estimate of s is available. We discuss estimation of s below.

We do not believe the simple SS-OS difference can be used to identify h^2 for labor market outcomes, however, as we expect the family environment relevant for labor market outcomes may differ for boys and girls in the same family. Some families are gender egalitarian, promoting sons' and daughters' labor market success to a similar degree, while others take more traditional views of gender roles, and may promote sons' success more than daughters'. This would violate the equal environments assumption and reduce the OS correlation relative to the SS correlation.

We extend the ACE model by adding a second family environment term, G, that varies by child's sex $g(i)$: $Y_i = A_i + C_{f(i)} + G_{f(i)g(i)} + E_i$. We assume that for two siblings *i* and *j* in the same family f, $corr\left(G_{fg(i)}, G_{fg(j)}\right) = 1$ if $g(i) = g(j)$ but $corr\left(G_{fg(i)}, G_{fg(j)}\right) = \tau < 1$ if $g(i) \neq g(j)$. For simplicity, we assume $cov(A,G) = cov(C,G) = 0$, though this is not essential. In this extended model:

$$cov_{MZ} = V(A) + V(C) + V(G) + 2cov(A,C)$$
$$cov_{DZSS} = R_A V(A) + V(C) + V(G) + 2cov(A,C)$$
$$cov_{SS} = (s + (1-s)R_A)V(A) + V(C) + V(G) + 2cov(A,C)$$
$$cov_{OS} = R_A V(A) + V(C) + \tau V(G) + 2cov(A,C)$$

In the presence of a family component that varies by sex, we cannot isolate $V(A)$ based only on the difference in covariances between SS and OS twins, since:

$$cov_{SS} - cov_{OS} = s(1-R_A)V(A) + (1-\tau)V(G).$$

To proceed, we augment this system with covariances for non-twin siblings. We assume the G correlations are the same as for twins: 1 if same-sex and τ if opposite sex. But the common family environment effect C may differ for siblings, whose time in the home only partly

overlaps. We assume that the correlation of this component is $\theta < 1$ for non-twin siblings. We further assume that A is correlated the same for siblings as for DZ twins, R_A. This implies

$$cov_{SSsib} = R_A V(A) + \theta V(C) + V(G) + 2cov(A, C)$$
$$cov_{OSsib} = R_A V(A) + \theta V(C) + \tau V(G) + 2cov(A, C)$$
$$cov_{SSsib} - cov_{OSsib} = (1 - \tau)V(G).$$

Combining the expressions for the differences in covariances between same-sex and opposite sex twins versus siblings, we obtain:

$$h^2 = \frac{(cov_{SS} - cov_{OS}) - (cov_{SSsib} - cov_{OSsib})}{s(1 - R_A)V(A)}.$$

The numerator of this expression is the difference in differences in covariances between same sex and opposite sex twins versus siblings. The denominator is a scale factor that reflects the fraction of MZ's among same sex twins (s) and the share of the genetic component shared by DZ twins (R_A). Note that this expression does not depend on the specific value of θ.

We assume $R_A = 0.5$ for the estimates presented in the text. Existing research suggests that assortative mating may lead to a positive correlation between husbands and wives in the components of genes that are related to education, suggesting a higher value of R_A. For example, Hugh-Jones et al. (2016) report a correlation of 0.14 in the polygenic scores for education between spouses. The same correlation in the genetic components of earnings would imply $R_A = 0.57$, leading to estimates of h^2 that are 16% above our reported values.[6] We also assume that the variances of all elements of our extended ACE model are constant across types of siblings (twins vs. not, SS vs. OS), and report estimated correlations and the value h^2 rather than $V(A)$.

To estimate $s \equiv p(MZ|SS)$, we first use the excess fraction of SS twins above the rate for DZ twins to estimate the share of MZ twins:

$$p(MZ) = \frac{p(SS) - p(SS \mid DZ)}{1 - p(SS \mid DZ)}$$

[6] If assortative mating leads to spouses' genetic indices being correlated 0.14, the correlation between dizygotic siblings' indices is 0.5*(1+0.14) = 0.57.

Assuming that the sexes of DZ twins are determined independently with probability π of being male (which we estimate at 0.508), we get $p(SS|DZ) = \pi^2 + (1-\pi)^2 = 0.500$. With $p(SS) = 0.708$ (in our sample), we estimate $p(MZ) = 0.416$ and $p(MZ|SS) = 0.588$.

A key assumption of our E-ACE model is that the gender-specific component of family environment (G) is perfectly correlated across siblings and additively separable from the non-gendered component. This allows us to remove the bias in $cov_{SS} - cov_{OS}$ by subtracting off the analogous difference for siblings. A second assumption is that the correlation between the shared family component (C) in sibling outcomes is invariant to the age gap between siblings. If instead the correlation decays with the age gap, but at the same rate for SS and OS siblings, then our approach remains valid provided the distributions of age gaps are similar for SS and OS siblings. The parallel and nearly flat profiles of the correlations in log earnings for SS and OS siblings at different age gaps shown in Figure 3 suggest that the decay rates of the sibling correlations are similar for SS and OS siblings and very small.

Standard errors

We use a heteroskedasticity-robust regression-based approach to compute the standard errors of our correlations. If β_{XY} is the coefficient of a regression of *Y* on *X*, it is related to the correlation coefficient by $\rho_{XY} = \frac{\sigma_X}{\sigma_Y}\beta_{XY}$, where σ_Y and σ_X are the standard deviations of *Y* and *X*, respectively. We thus compute the sampling variance of the correlation coefficient as $V(\rho_{XY}) = \frac{\sigma_X^2}{\sigma_Y^2}V(\beta_{XY})$, where $V(\beta_{XY})$ is the heteroskedasticity-robust variance of β_{XY} and we treat the variances of *X* and Y as known. This yields slightly larger standard errors than the traditional formula for the standard error of an estimated correlation, $\frac{1-\rho^2}{\sqrt{N}}$.

Our estimate of h^2 is computed from the difference-in-differences of correlation coefficients from independent samples, multiplied by a constant $\frac{1}{s(1-R_A)}$. We treat $s(1-R_A)$ as known. Thus, the variance of h^2 is the sum of the variances of the underlying correlations, multiplied by the square of the constant.

Decomposing log earnings using the AKM model

We use a decomposition based on the method described by Abowd et al. (1999) to decompose quarterly earnings in the LEHD file. In brief, the model can be written as:

$$y_{it} = \alpha_i + \psi_{j(i,t)} + X_{it}\beta + \xi_{it},$$

where y_{it} represents the log earnings of individual i in quarter t, α_i is a person fixed effect, $\psi_{j(i,t)}$ is a fixed effect for the establishment at which the person works in quarter t, $j(i,t)$ is an index function giving the workplace of individual i in quarter t, $X_{it}\beta$ captures effects of time-varying observable covariates, such as experience and calendar time, and ξ_{it} is assumed to be orthogonal to the sequence of establishment effects, conditional on α_i (see Card et al., 2013). In the AKM literature α_i is interpreted as an individual's earnings capacity at any job, while ψ_j is interpreted as a wage premium or discount at establishment j relative to pay in a reference workplace (which we take to be the average employer in the restaurant sector).

We estimate this model using data on all individuals in the LEHD from 2018Q1 to 2023Q1, omitting data from 2020Q2 through 2020Q4 during the COVID epidemic. The sample is limited to people who have at least 8 quarters with earnings over $3,800. We use only quarters when an individual is between 20 and 60 years of age and has a single employer. The first and last quarters on a given job are also excluded. After model estimation we assign the estimated person effects to the individuals who are either twins or siblings in our analysis sample. Similarly, we assign the mean of the estimated pay premiums for firms at which they worked in all the quarters they are observed satisfying the restrictions for our estimation sample. The sum of these two components accounts for 82% of the variance in person-quarter log earnings in the cohorts from which our twin sample is drawn (Card, Rothstein, and Yi, 2026). We further decompose the firm pay premiums into location, industry, and residual components, following methods developed in Card, Rothstein, and Yi (2024a).

We note that the AKM model makes the strong assumption that the person and firm components of wages *multiply* together, so the logarithm of earnings is additive in the two parts. Violation of this assumption could lead to problems in interpreting our decomposition. Fortunately, tests reported by Card et al. (2013), Card et al. (2025), and Lachowska et al. (2026)

and summarized in Kline (2024) suggest that violations of additivity in the logarithm of earnings, while present, are relatively minor.

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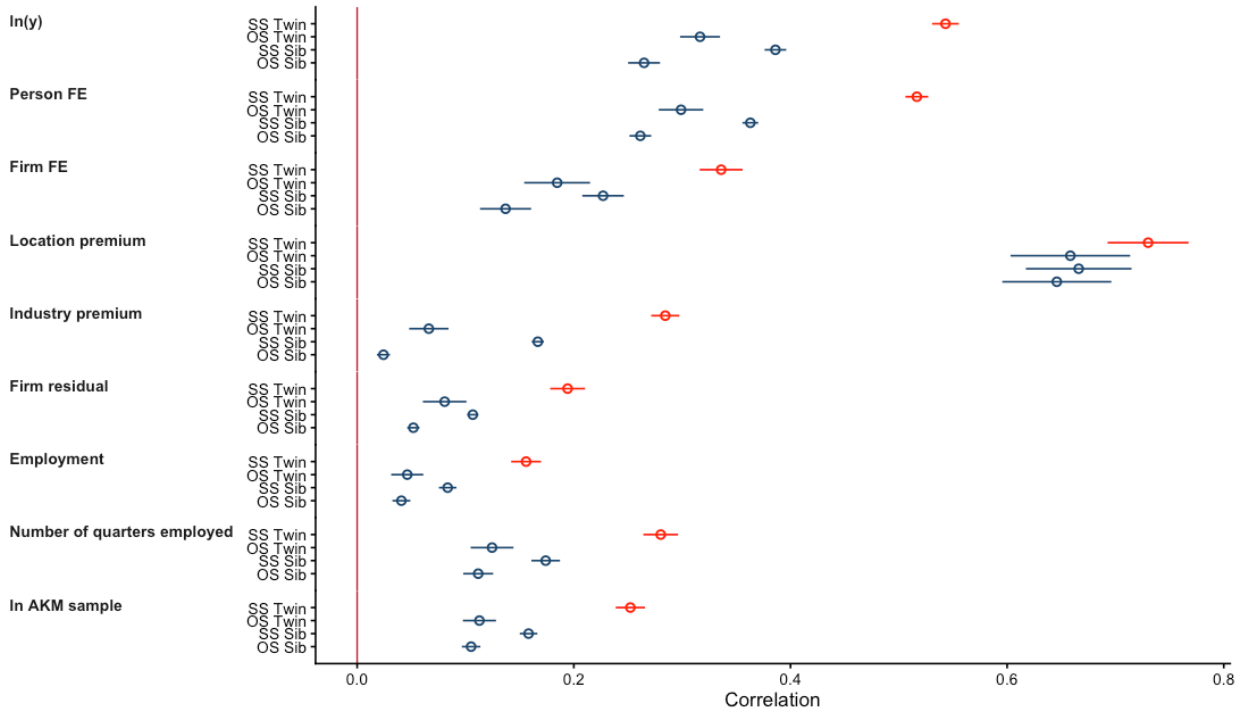
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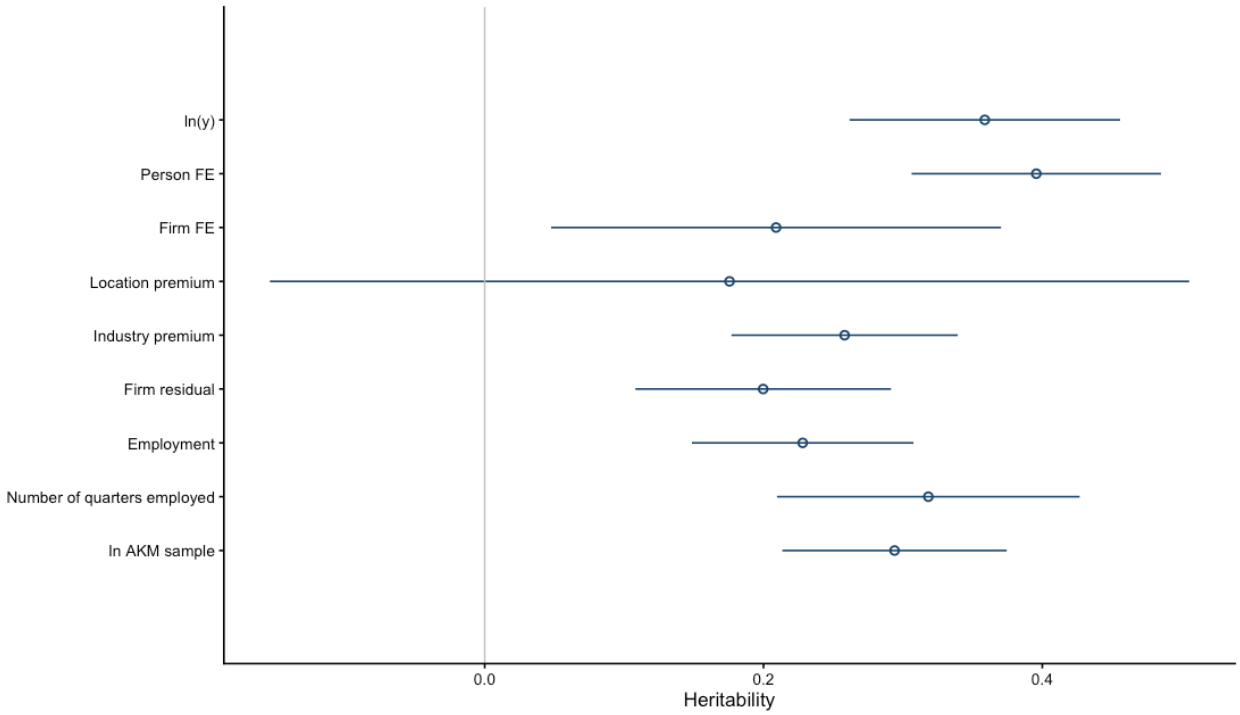
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Figure 1. Between-sibling and between-twin correlations, same-sex (SS) and opposite-sex (OS)



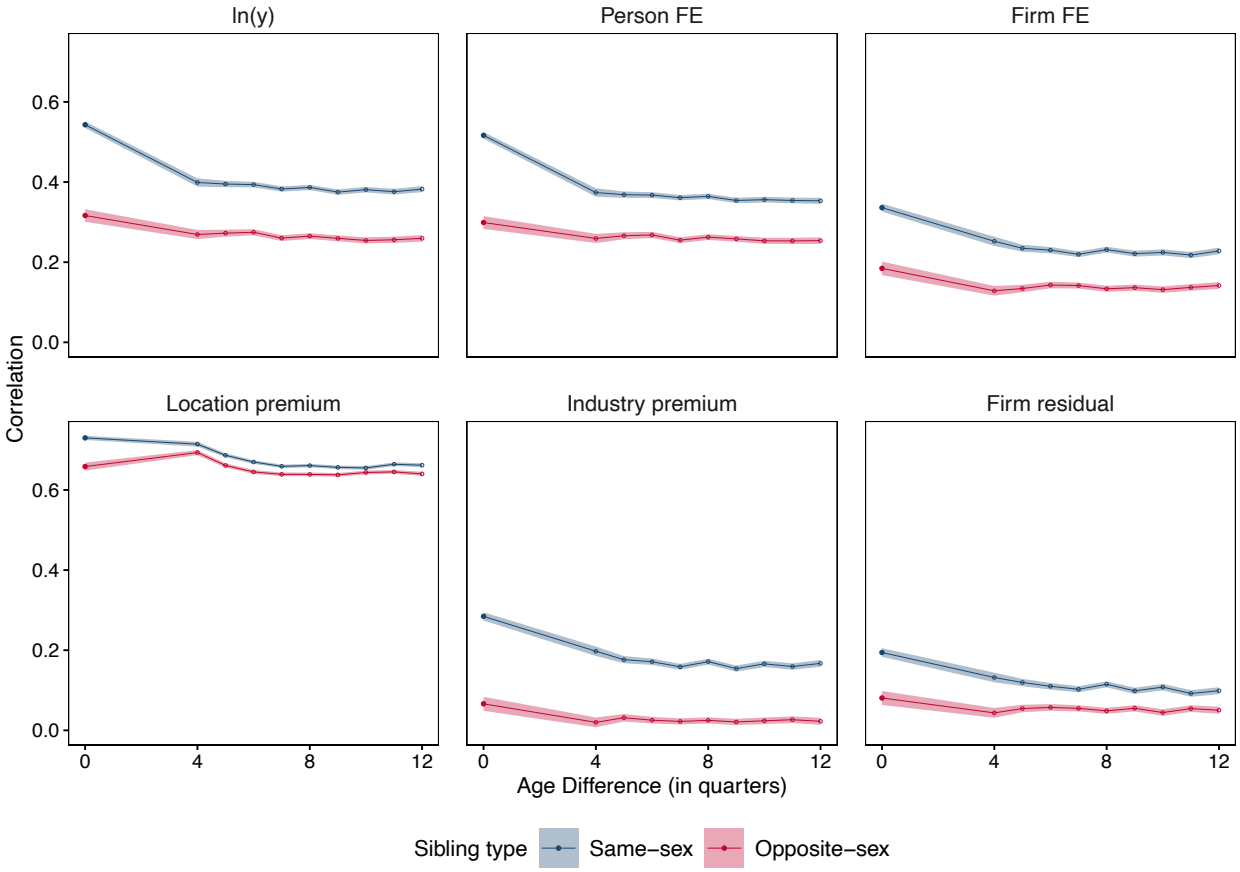
Notes: Horizontal lines represent the 95% confidence intervals. Log earnings and its components, presented in the first six panels, are computed over the sample of twin and sibling pairs who are both stably employed, as described in the data section. Employment outcomes, in the final three panels, are computed over all twin pairs, including those who are not both stably employed.

Figure 2. Heritability estimates from E-ACE model



Notes: Horizontal lines represent the 95% confidence intervals.

Figure 3. Between-sibling correlations, by age difference between siblings



Notes: Shaded region indicates 95% pointwise confidence interval. This is computed using traditional formula for the standard error of a correlation coefficient.

Table 1. Correlations of outcomes among twins and siblings

	Correlation				Difference, SS -	Difference, SS twins	Difference-in-	Implied
	SS twins	OS twins	SS siblings	OS siblings	OS twins	vs. SS siblings	differences	heritability
A. Correlations/differences								
Employment outcomes								
Any employment	0.156	0.046	0.084	0.041	0.110	0.072	0.067	0.23
No. of quarters employed	0.280	0.125	0.174	0.112	0.156	0.106	0.093	0.32
In high-employment sample	0.252	0.113	0.158	0.105	0.139	0.094	0.086	0.29
Earnings outcomes								
Log earnings	0.543	0.317	0.386	0.265	0.227	0.157	0.105	0.36
AKM person effect	0.517	0.299	0.363	0.261	0.218	0.154	0.116	0.40
AKM firm effect	0.336	0.185	0.227	0.137	0.151	0.109	0.061	0.21
Location premium	0.730	0.658	0.666	0.646	0.072	0.064	0.052	0.18
Industry premium	0.284	0.066	0.167	0.024	0.218	0.118	0.076	0.26
Firm residual	0.194	0.081	0.107	0.052	0.113	0.088	0.059	0.20
B. Number of observations (rounded) and standard errors								
Employment outcomes: No. obs.	60,000	26,500	1,072,000	1,096,000				
Any employment	0.007	0.007	0.004	0.004	0.010	0.008	0.012	0.040
No. of quarters employed	0.008	0.010	0.007	0.007	0.013	0.010	0.016	0.054
In high-employment sample	0.007	0.008	0.004	0.004	0.010	0.008	0.012	0.040
Earnings outcomes: No. obs.	31,500	13,000	507,000	500,000				
Log earnings	0.006	0.009	0.005	0.007	0.011	0.008	0.014	0.049
AKM person effect	0.005	0.010	0.004	0.005	0.012	0.006	0.013	0.045
AKM firm effect	0.010	0.015	0.010	0.012	0.018	0.014	0.024	0.081
Location premium	0.019	0.028	0.024	0.025	0.033	0.031	0.048	0.165
Industry premium	0.006	0.009	0.003	0.003	0.011	0.007	0.012	0.041
Firm residual	0.008	0.010	0.003	0.003	0.013	0.008	0.013	0.046

Notes: SS is same-sex, OS is opposite sex. In panel A, entries in first four columns are estimated correlations in outcomes (given by row title) between twins or siblings as indicated in column heading. Entries in columns 5-7 are differences in correlations; column 8 is the estimated heritability. See text for definition of AKM person effects, AKM firm effects, and components of AKM firm effect. In panel B, entries correspond to number of observations (rounded to the nearest 500) or estimated standard errors of estimates in panel A.

Table 2. Sensitivity to firm overlap

	Estimated heritability / characteristics of subsample	
	Full sample	No firm Overlap
Estimates of heritability		
Log earnings	0.36	0.31
AKM person effect	0.40	0.35
AKM firm effect	0.21	0.07
Location premium	0.18	0.13
Industry premium	0.26	0.12
Firm residual	0.20	-0.01
Number of twins	44,500	40,500
Fraction in same firm	9%	0%
Fraction in same industry	31%	25%
Comparison: Siblings		
Fraction in same firm	4%	0%
Fraction in same industry	22%	19%
Standard errors		
Log earnings	0.05	0.051
AKM person effect	0.04	0.046
AKM firm effect	0.08	0.084
Location premium	0.16	0.170
Industry premium	0.04	0.041
Firm residual	0.05	0.044

Notes. "No firm overlap" sample in column 2 excludes twin or sibling pairs for which there is any overlap in the list of firms at which the two siblings worked in 2018-2023Q1. Standard errors for heritability statistics are computed from the standard errors of the correlations using the delta method.